Penguin Species Classification Memo

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Contents

1 Executive Summary

Dr. Sharma provided measurements from an unidentified penguin specimen with a bill length of 45 mm and a flipper length of 220 mm. Using the Palmer Penguins dataset, I compared these measurements to known Adelie, Chinstrap, and Gentoo penguins. Both visual comparison and statistical analysis suggest that the specimen is most likely a Gentoo penguin. The specimen's flipper length is particularly characteristic of Gentoo penguins. While the classification appears strong, confidence could be improved with additional measurements such as body mass or bill depth.

2 Background and Objective

The objective of this project is to identify the most likely species of an unlabeled penguin specimen discovered during a previous expedition. The available measurements are bill length (45 mm) and flipper length (220 mm). Dr. Sharma would like to determine:

1. The most likely species.
2. The confidence in that classification.
3. How the specimen compares visually to known penguin species.

3 Data Description

The Palmer Penguins dataset contains measurements from three penguin species: Adelie, Chinstrap, and Gentoo. For this analysis, bill length and flipper length were used because these measurements were available for the unknown specimen. Records with missing values were removed prior to analysis.

4 Exploratory Analysis

Figure 1 shows the relationship between bill length and flipper length for the three species. Gentoo penguins generally have longer flippers than Adelie and Chinstrap penguins. The unknown specimen falls near the Gentoo cluster due to its unusually long flipper length of 220 mm.

Table 1: Comparison of the Unknown Specimen with Typical Measurements by Species (mm)

species	Avg Bill Length	Avg Flipper Length
Adelie	38.8	190.0
Chinstrap	48.8	195.8
Gentoo	47.5	217.2
Unknown Specimen	45.0	220.0

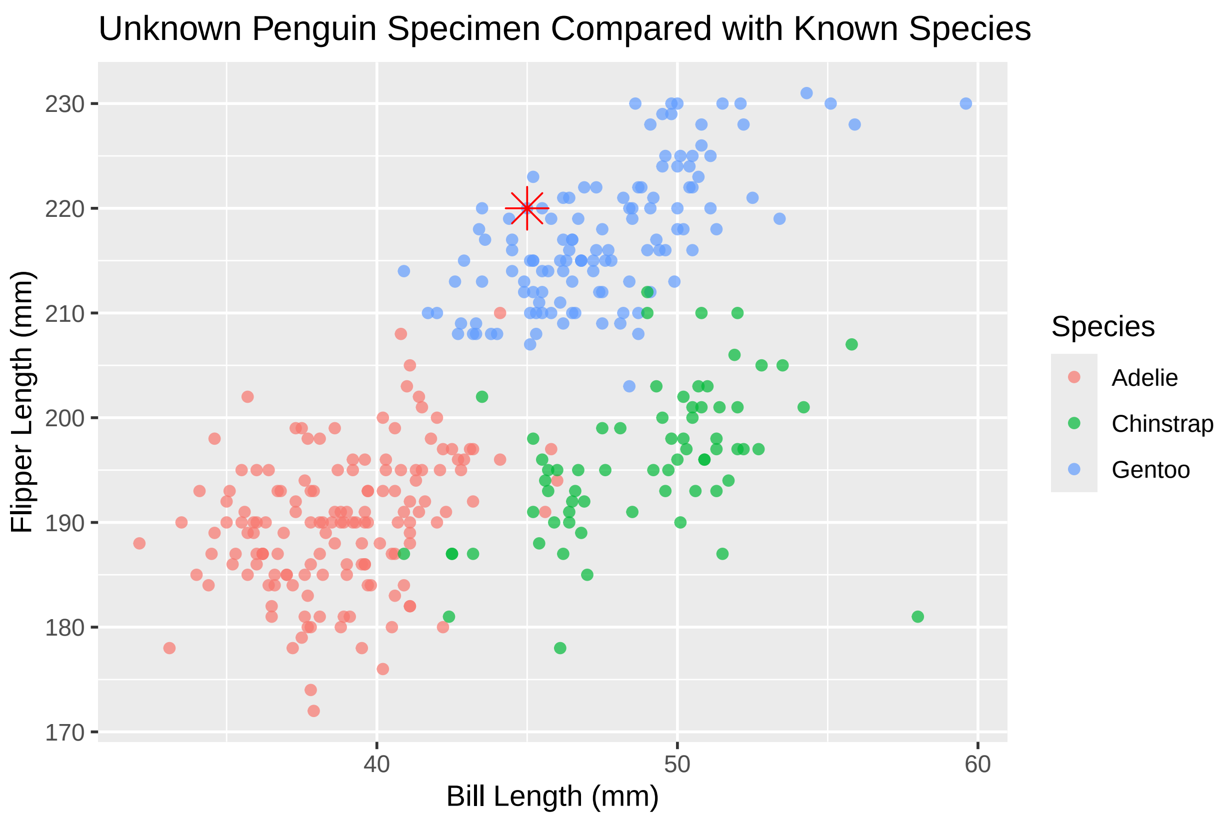

Figure 1. Scatterplot of bill length and flipper length by species.

5 Analytical Approach

The unknown specimen was compared with known penguin measurements using both visual comparison and statistical classification techniques.

6 Results

The estimated probabilities for each species are shown in Table 2. The species with the highest probability is considered the most likely classification.

Table 2: Estimated Probability of Species Membership

Species	Probability
Adelie	0
Chinstrap	0
Gentoo	100

The unknown specimen has a bill length of 45 mm and a flipper length of 220 mm. While its bill length falls between the Chinstrap and Gentoo averages, its flipper length is most consistent with Gentoo penguins, providing strong evidence for that classification.

7 Interpretation for Dr. Sharma

Based on bill length and flipper length, the unknown specimen is most likely a Gentoo penguin. The strongest evidence is its long flipper length of 220 mm, which visually places it closest to the Gentoo cluster. Both the comparison table and scatterplot place the specimen closest to the Gentoo species. The statistical analysis further supports this conclusion by assigning the highest probability to Gentoo among the three candidate species.

8 Limitations

This classification is based only on two measurements: bill length and flipper length. Including body mass, bill depth, sex, island, or other biological information could improve confidence. Because the specimen is preserved and from a prior expedition, measurement error or preservation effects may also affect the result.

9 Recommendation

Based on both visual inspection and statistical classification, the specimen should be provisionally classified as a Gentoo penguin. The evidence supporting this classification is strong because the specimen's flipper length is highly characteristic of Gentoo penguins. If additional measurements such as body mass or bill depth become available, the classification should be re-evaluated to further increase confidence.